

Production and Costs

Module 3

Management 405 · EMBA

Prof. Nico Voigtländer · UCLA Anderson

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General Logistics

- Coffee & Econ on Sunday Nov 8 at 9am
  - In the class calendar
  - Send me questions / suggestions

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Midterm Logistics: Timing, Coverage, and Help

- Open Fri 10/30, 12:01 am – Sat 10/31, 11:59 pm
  - 3 hours from first opening; the clock cannot be stopped
  - Latest full-length start: Sat 10/31, 8:59 pm
- TA on Zoom: Sat 10/31, 9 am – noon and 1 – 4 pm
  - Join from your phone before starting; raise your hand for a breakout room
  - Phone only for the TA call
- Material: Modules 1 – 3 (incl. Problem Sets 1 + 2)
  - Problem-solving exercises similar to the problem sets
- Practice Exam on the class website (under Week 6)

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Midterm Rules: LockDown Browser, Paper Only, One PDF

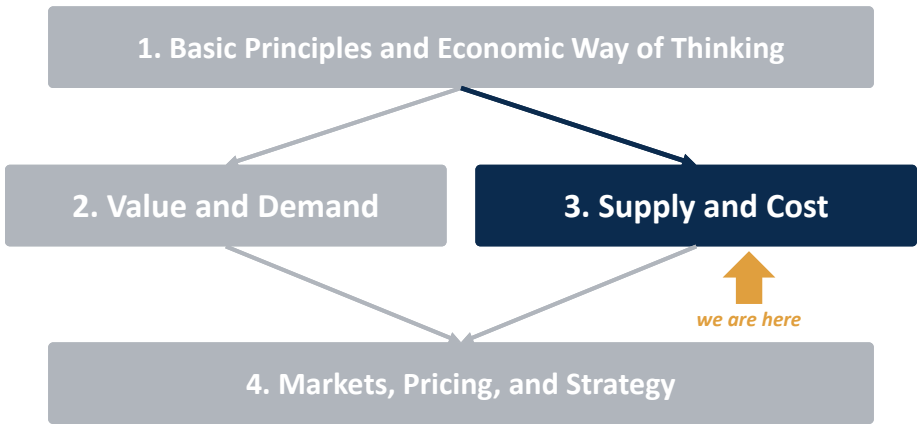
- Respondus LockDown Browser + webcam required
  - Go over the “Lockdown Browser Practice” first
  - Show your workspace at the start and after every break
- Allowed: anything on paper. Not allowed: other devices, internet, AI
  - Phone only for the TA call; the calculator is built into the browser
- Part I (Multiple Choice): pick one answer. Part II: type final answers in the boxes
  - Show your work on paper for partial credit; email the TA about extra assumptions
- After the exam: webcam photos, then upload ONE PDF within 15 minutes
  - Only pages shown on the webcam count

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Module 3
Some Applications of the Material Covered in the Videos

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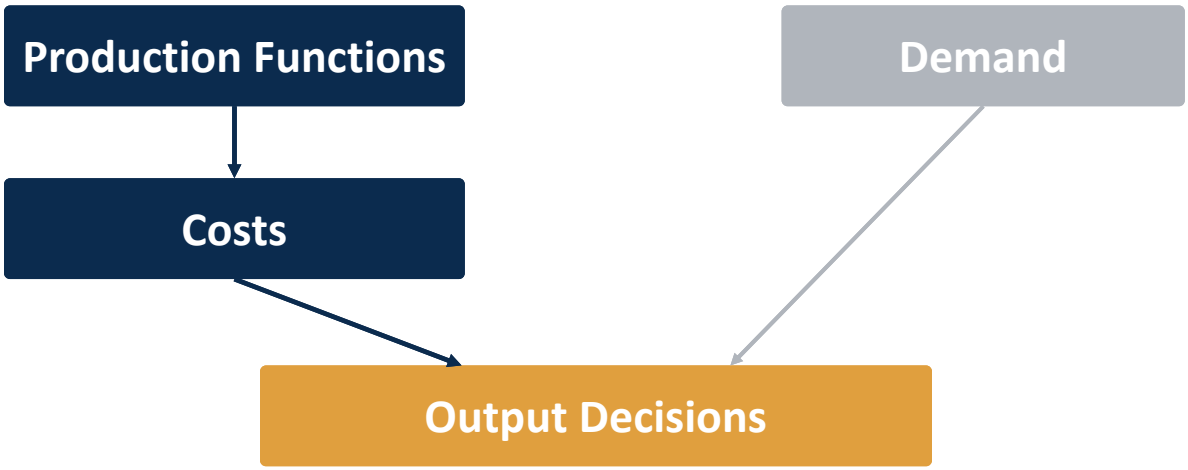
Agenda for the Class



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Module 3 · Video 1 · Big Picture

Big Picture of Module 3



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Module 3 · Video 1 · Agenda

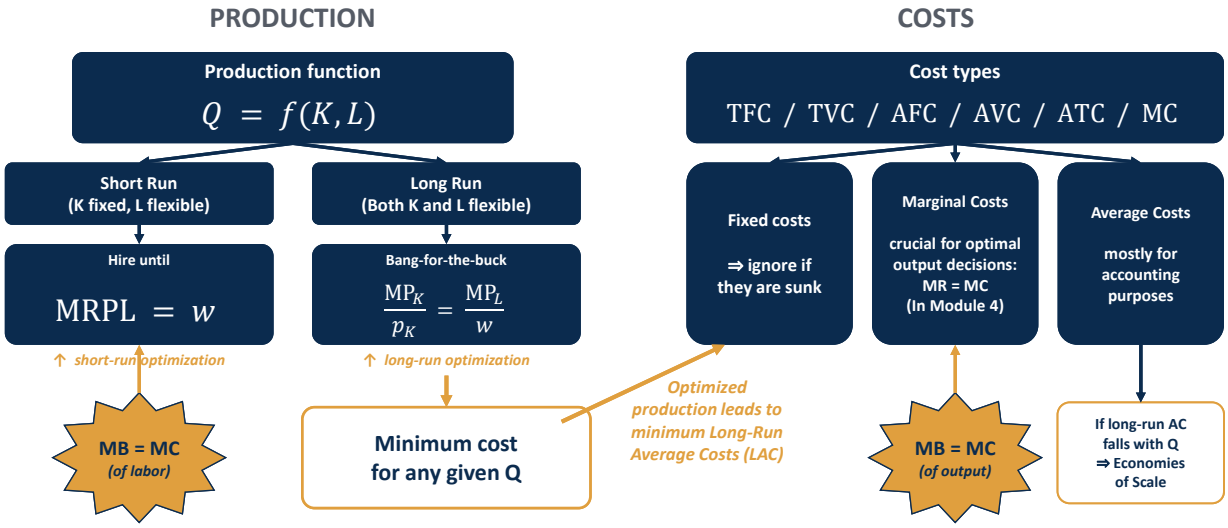
Outline of Module 3

1. **Introduction to Module 3** — Video 1
   A brief overview of what we cover in Module 3
2. **The Production Function** — Video 2
   How inputs (capital and labor) are turned into output
3. **Short Run: Hiring Decisions** — Video 3
   How many workers to hire when capital is fixed
4. **Wage Searchers** — Video 4
   When hiring one more worker bids up the wage for everyone
5. **Long Run: The Optimal Input Mix** — Video 5
   Choosing between capital and labor when both can be adjusted
6. **Cost Concepts** — Video 6
   Fixed, variable, marginal, and sunk costs, and which ones matter
7. **Economies of Scale and Scope** — Video 7
   When bigger or broader production lowers cost per unit

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Module 3 · Concept Map

How the Pieces of Module 3 Connect



Module 3 · Video 3 · Short Run: Hiring Decisions

The Optimal Hiring Rule in the Short Run

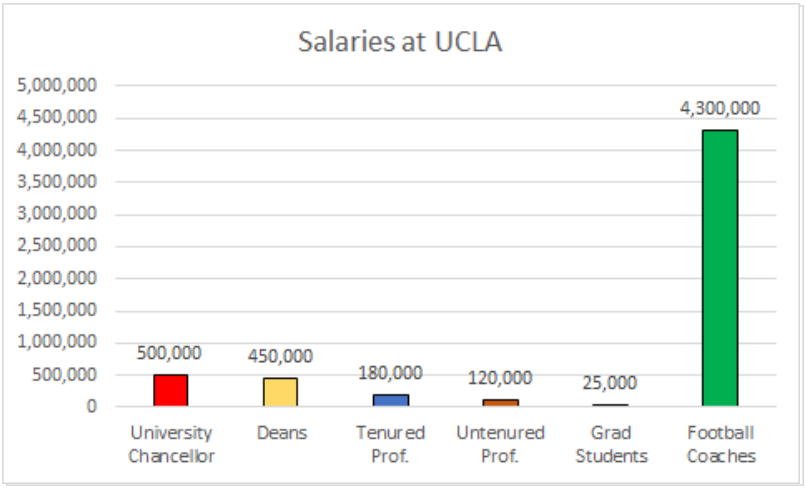
- Recall: MRPL: Marginal Revenue Product of Labor
  - By how much revenues go up if we hire one more worker
- If MRPL > wage: Hire more labor

Optimal Number of Workers: L* where: MRPL = w ← MB = MC (of labor)

- Note:
  - This holds in the short run, when K is fixed while labor is the flexible input

Module 3 · In Class · Examples · Wage Searchers

Are Real-World Wages = MRPL?



Search tool: https://ucannualwage.ucop.edu/wage/

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Module 3 · Video 4 · Agenda

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1	Introduction to Module 3	Video 1
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4	**Wage Searchers** When hiring one more worker bids up the wage for everyone	Video 4
5	Long Run: The Optimal Input Mix	Video 5
6	Cost Concepts	Video 6
7	Economies of Scale and Scope	Video 7

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Example: The Full Cost of Poaching an AI Researcher

- Anthropic is trying to poach a star researcher from Google DeepMind
  - She would join Anthropic for a $5M annual salary
- Anthropic already employs 2 star researchers, each earning $3.5M
  - If the new researcher is hired, the existing two will demand the same salary

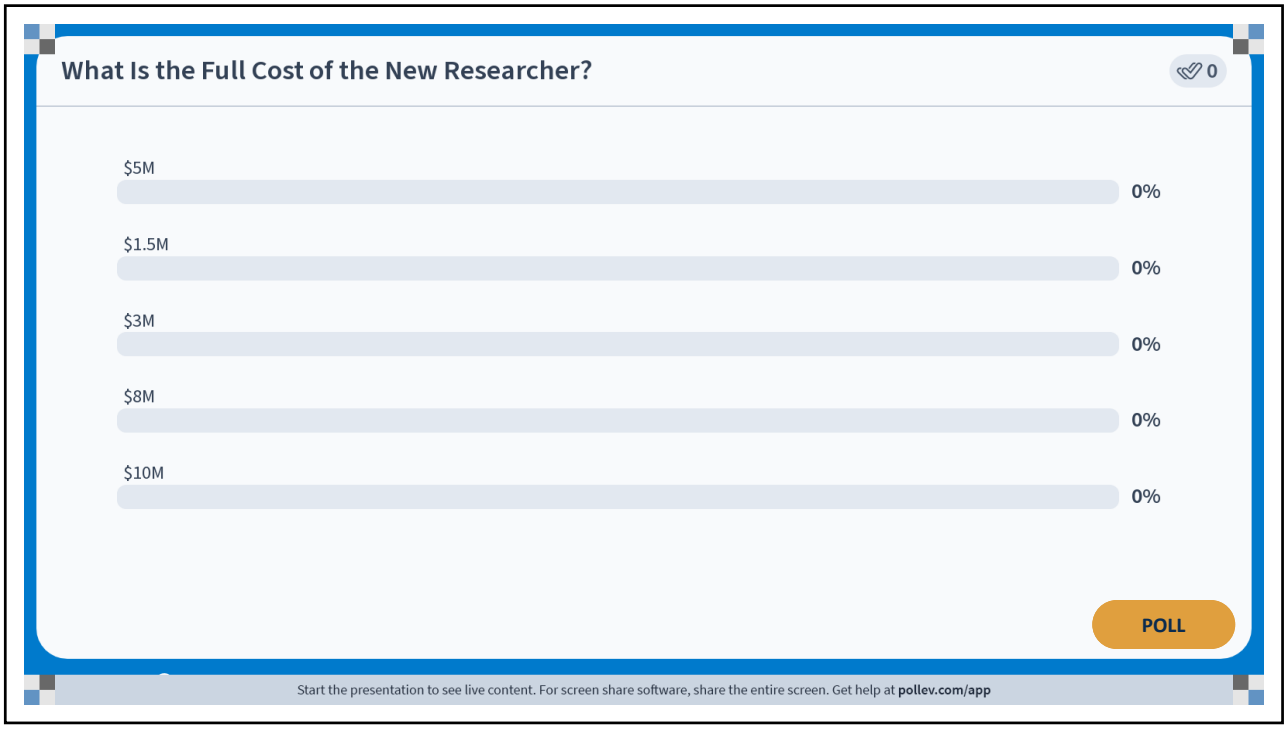
→ **What is the marginal cost of the 3rd researcher?**



Demis Hassabis (CC BY, C. Michel via Wikimedia)

Poll Break

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Solution

Poll Break

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Optimal Combination of Inputs

- Optimal Input Mix: How much K and how much L?

Decision rule for the long run:

$$\frac{MP_K}{p_K} = \frac{MP_L}{w}$$

MP_K : Marginal Product of Capital
MP_L : Marginal Product of Labor
p_k : Price of Capital
w : Wage

Bang for the Buck Rule

- Holds when both L and K are flexible
- Refers to a given output quantity Q
- Assumes input prices w and p_k are constant

Teaching Note: Bang-for-the-Buck Rule

"Bang for the Buck" in Grocery Shopping

- P&G offers two sizes for its flagship detergent
  - Small Tide Ultra Oxi (29 loads) for $17
  - Large Tide Ultra Oxi (81 loads) for $35

$$\frac{MP_{small}}{p_{small}} = \frac{29}{17} = 1.7$$

$$\frac{MP_{large}}{p_{large}} = \frac{81}{35} = 2.3$$



$$\frac{MP_{small}}{p_{small}} < \frac{MP_{large}}{p_{large}}$$



Bang-for-the-buck: a universal decision rule

Applying the 'Bang for the Buck' Rule

('recipe' for exams)

How to adjust L and K in order to produce a given quantity at lower cost

- If $\frac{MP_L}{w} > \frac{MP_K}{p_K}$ → use more *L* and less *K*
  - Decreasing marginal returns imply:
    - Due to increased *L*, MP_L will fall as *L* increases
    - Due to reduced *K*, MP_K will rise as *K* decreases
    - Thus, the two fractions move closer to each other
- If $\frac{MP_L}{w} < \frac{MP_K}{p_K}$ → use less *L* and more *K*
- …continue to adjust until $\frac{MP_L}{w} = \frac{MP_K}{p_K}$

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Fixed, variable, marginal, and sunk costs, and which ones matter

7 Economies of Scale and Scope — Video 7

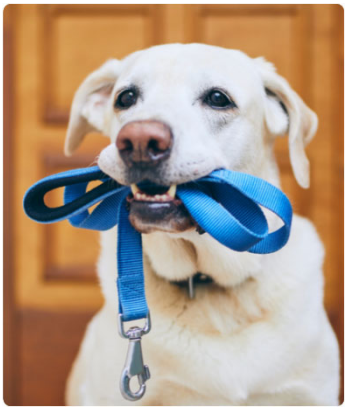
Three Cost Types

Variable Costs	Fixed Costs	Sunk Costs
Depend on volume produced (Q) → Always consider!	Do not depend on quantity produced (Q) → Consider for long-run decisions (entry, exit, capacity)	A fixed cost that cannot be recovered → Ignore!
Examples • Buy raw materials • Hire workers • Shipping & packaging	***Examples*** • Rent / lease payments • Insurance & property tax • R&D investments (for future innovation)	***Examples*** • Past R&D • Past advertising • Non-refundable deposits

Sunk costs should not affect managerial decisions

Cost Example: Dog Leashes

- You produce dog leashes
  - Already bought: lots of nylon yarn and a webbing loom (cannot be resold)
  - You pay rent for the shop and employ sewing workers
- Last month:
  - Sold 1,000 leashes at P = \$40
  - TC = \$35,000
- Expected: you could sell 1,200 leashes at P = \$35
- Should you cut the price to \$35?



Which cost information do you need to make a decision?

Group Discussion

Solution

Group Discussion

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Relationship to Accounting: Ross Stores Annual Report

	Year Ended January 29, 2022	Year Ended January 30, 2021
Sales	18,916,244 $	12,531,565 $
Costs and Expenses		
Cost of goods sold ≈ TVC	13,708,907	9,838,574
Selling, general and administrative Mix FC & VC	2,874,469	2,503,281
Interest expense (income), net Part of FC	74,328	83,413
Total costs and expenses ≈ TC	16,657,704	12,425,268

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Why Studios Finish Movies They Know Lose Money

Waterworld (1995)



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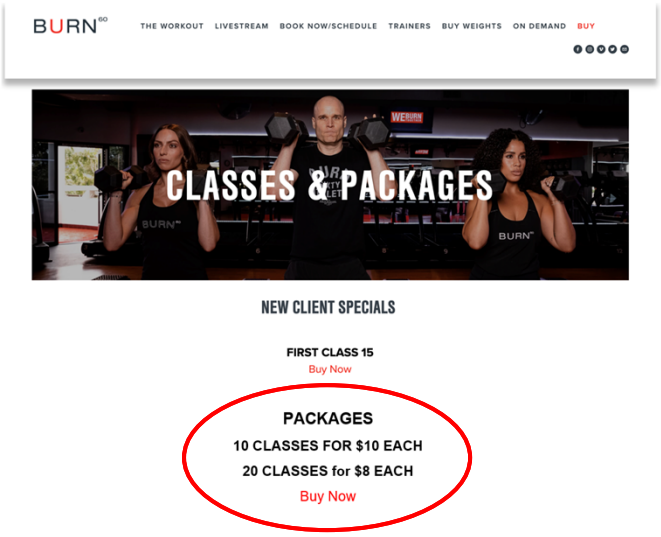
Waterworld: Sunk Costs and Decisions Over Time

	June 1994	September 1994	December 1994
Expected Revenues ($M)	150	150	150
Sunk Cost ($M)	16	100	140
Expected Additional Cost ($M)	84	40	35
Overall Cost (incl. sunk) ($M)	100	140	175
Expected Profit ($M)	+50	+10	−25
Decision	**Make the film!**	**Make the film!**	**Make the film!**

Ignore sunk costs — continue when Expected Revenue > Expected Additional Cost

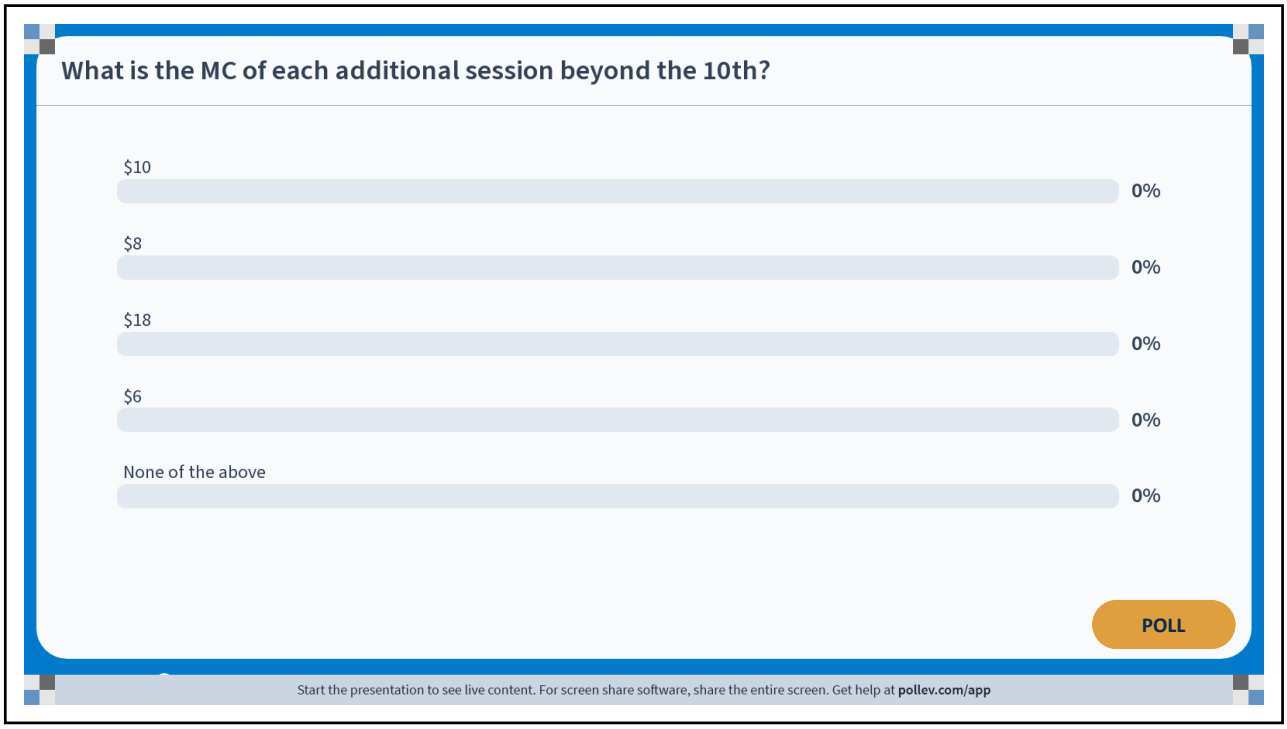
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Marginal Cost in Action: What Is the MC of a Burn60 Workout?



Poll Break

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Solution

Poll Break

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Marginal Cost in Finance: The True Rate on a Bigger Loan

Option 1		Option 2	
$100k loan		$110k loan	
at 5% annual interest		*at 6% annual interest*	
% Annual interest	$5,000	% Annual interest	$6,600
≡ Total loan amount	$100,000	≡ Total loan amount	$110,000
$ Annual interest cost	$5,000	$ Annual interest cost	$6,600

Extra $10k

What is the marginal interest rate on the additional $10k?
($6,600 – $5,000) / $10,000 = 16%

Marginal cost (16%) is much higher than average rate (6%)

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Amazon: Economies of Scale, Scope, or Both?

Jeff Bezos · 1997 interview



Group Discussion

Solution

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